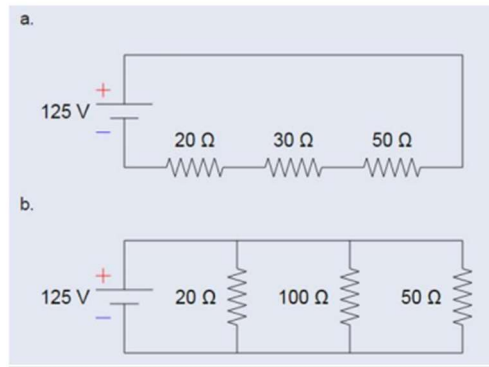


Part A



For each of these circuits:

1. Calculate the total resistance in ohms,
2. Calculate the total current from the power supply,
3. Calculate the current through each resistor
4. Calculate the Voltage drop through each resistor
5. Calculate the Power dissipated through each resistor (*use Watt's Law: Power = Voltage * Current*)

Circuit A:

1. Total $\Omega = R_1 + R_2 + R_3 = 20 + 30 + 50 = 100 \Omega$
2. $I = V/R = 125 / 100 = 1.25A$
3. Current is the same in series, thus each resistor has 1.25A going through it
4. Voltage drop: $V = IR$
 $20 \Omega = 1.25A * 20 = 25V$
 $30 \Omega = 1.25A * 30 = 37.5V$
 $50 \Omega = 1.25A * 50 = 62.5V$
5. Power: $P = VI$, but $V = IR$, hence $P = IRI = I^2R$
 $20 \Omega = 1.25^2 * 20 = 31.25W$
 $30 \Omega = 1.25^2 * 30 = 46.88W$
 $50 \Omega = 1.25^2 * 50 = 78.13W$

Circuit B:

1. Total $\Omega = (1/R_1 + 1/R_2 + 1/R_3)^{-1} = (1/20 + 1/100 + 1/50)^{-1} = 0.08^{-1} = 12.5\Omega$
2. $I = V/R = 125 / 12.5 = 10A$
3. Voltage is the same in parallel, so each resistor has 125V through it ($I = V/R$)
 $20\ \Omega = 125V / 20 = 6.25A$
 $100\ \Omega = 125V / 100 = 1.25A$
 $50\ \Omega = 125V / 50 = 2.5A$
4. Parallel circuit = Same voltage drop for all: 125V
5. V is consistent now in parallel, so we use $P = VI$
 $20\ \Omega = 125 * 6.25 = 781.25W$
 $100\ \Omega = 125 * 1.25 = 156.25W$
 $50\ \Omega = 125 * 2.5 = 312.5W$